Name _____ Student ID _____

Part I: Multiple Choice (20 points)

Circle the one correct answer in each question.

1. What is a trap in an operating system?

   (a) A queue for blocked processes
   (b) A synchronous transfer to an OS handler, as used for a system call
   (c) An interrupt generated only by an external device
   (d) A page selected for replacement

2. What could happen without separation between user mode and privileged mode?

   (a) The OS could no longer create any child process
   (b) All memory accesses would automatically become safe
   (c) A process could directly access protected data or devices
   (d) Every process would have to remain blocked

3. In UNIX, which numeric identifier identifies a live process on the system?

   (a) Process Control Block
   (b) Device queue
   (c) Process Identifier (PID)
   (d) Page-frame number

4. Which register holds the address of the next instruction to execute?

   (a) A general-purpose data register
   (b) The program counter
   (c) The stack pointer
   (d) The page-table base register

5. Which OS concern protects one application from unauthorized access by another?

   (a) Mobility
   (b) Throughput
   (c) Isolation
   (d) Convenience

6. Which listed facility is required specifically for voluntary process switching in a non-preemptive system?

   (a) A timer that periodically forces a context switch
   (b) A TLB
   (c) A cache
   (d) None; cooperative switching does not require timer-driven preemption

7. Which part of a process is normally maintained in kernel space?

   (a) The user stack
   (b) The process descriptor
   (c) The user text segment
   (d) The user heap

8. In the conventional address-space layout, the heap grows upward. Which region typically grows in the opposite direction?

   (a) The text segment
   (b) The static data segment
   (c) The stack
   (d) The executable file header

9. In the basic three-state model, when can a ready process move directly to blocked?

   (a) When it reads a file before being dispatched
   (b) Whenever a timer interrupt occurs
   (c) Whenever the ready queue is nonempty
   (d) It cannot; it must first run and request an event or I/O

10. The state of a process is defined by which of the following?

    (a) Its final activity
    (b) The activity it completed previously
    (c) The activity it will perform next
    (d) Its current activity

11. A process is blocked on a disk read. What is it waiting for?

    (a) The disk I/O to complete
    (b) Its next time quantum to expire
    (c) The OS to delete its executable file
    (d) Its priority to become the highest

12. What does it mean when a process is a zombie?

    (a) It is running and cannot be preempted
    (b) It has exited, but its exit status has not yet been reaped
    (c) It is ready but has never been dispatched
    (d) It is waiting indefinitely for disk I/O

13. Which data structure keeps a process's state, saved registers, and scheduling information?

    (a) A translation lookaside buffer
    (b) A user stack alone
    (c) A process control block
    (d) A page frame

14. A child prints before exiting. Its parent successfully waits for that child and then prints. Which order is guaranteed?

    (a) The parent's print precedes the child's print
    (b) Both prints occur simultaneously
    (c) The child restarts at the beginning of main()
    (d) The child's print precedes the parent's print

15. Which non-preemptive policy executes ready jobs in their arrival order?

    (a) FIFO (FCFS)
    (b) SJF
    (c) STCF
    (d) Priority scheduling

16. Starting with one process, every process executes two successful calls: fork(); fork(); then prints once. How many prints occur in total? Assume no earlier buffered output.

    (a) One
    (b) Two
    (c) Three
    (d) Four

17. Which statement about standard MLFQ is NOT true?

    (a) A new job enters at the lowest priority
    (b) Different queues can have different priorities
    (c) Jobs at the same priority can use RR
    (d) A ready job belongs to one priority queue at a time

18. Which statement about virtual memory is NOT true?

    (a) Processes need protection from one another
    (b) A virtual address is translated to a physical address
    (c) Physical memory must exceed the combined virtual address spaces
    (d) Backing storage can hold pages that are not resident

19. Which statement describes CPU virtualization provided by a modern operating system?

    (a) Each process appears to have its own CPU
    (b) A single CPU core physically executes all processes at once
    (c) A CPU executes user instructions directly from disk
    (d) Every process must always have the same program-counter value

20. Which statement about memory allocation is true?

    (a) Variable-size partitions cannot suffer external fragmentation
    (b) Fixed-size partitions can suffer internal fragmentation
    (c) Compaction removes unused space inside every allocated block
    (d) Single-partition allocation always uses memory best

21. What does a CPU process-selection algorithm select?

    (a) A virtual page to bring into RAM
    (b) A program file to remove from disk
    (c) A ready process to run next
    (d) A physical frame to evict

22. What does a page-replacement algorithm select?

    (a) A resident page to evict when a frame is needed
    (b) A process to execute next
    (c) An instruction to fetch from the ready queue
    (d) A new virtual address for the program counter

23. Which of the following calls does not return to the original calling point when it succeeds?

    (a) fork()
    (b) exec()
    (c) wait()
    (d) None of these

24. What is a disadvantage of basic segmentation?

    (a) Every segment must have the same fixed size
    (b) It can suffer external fragmentation
    (c) It cannot provide bounds checking
    (d) It eliminates all memory-management overhead

25. What does demand paging mean?

    (a) Loading all pages before the process starts
    (b) Keeping every page permanently in physical memory
    (c) Bringing in a page when it is referenced and not resident
    (d) Preloading only pages explicitly named by malloc()

26. Which is NOT a normal step in handling a legal, not-present page fault?

    (a) Locate the page on backing storage
    (b) Obtain a frame, replacing a page if necessary
    (c) Update the page table after the page is read
    (d) Always terminate the process without attempting a page-in

27. Which pair is used to relocate and bounds-check an address in a segment?

    (a) Virtual page number and page-fault count
    (b) Time quantum and process priority
    (c) Physical frame number and CPU burst time
    (d) Segment base address and segment limit

28. For a fixed virtual-address-space size and fixed PTE size, what is an effect of increasing the page size?

    (a) A smaller single-level page table
    (b) More virtual pages
    (c) Fewer page-offset bits
    (d) External fragmentation between fixed-size frames

29. Which scheduling policy does not require timer-driven preemption?

    (a) STCF
    (b) RR
    (c) Preemptive MLFQ
    (d) Non-preemptive FIFO

30. Where can virtual-memory pages be kept when they are not resident in RAM?

    (a) Only in the CPU registers
    (b) On backing storage such as a disk
    (c) Only in the TLB
    (d) Inside the page-table entries themselves

31. What is response time in CPU scheduling?

    (a) Completion time minus arrival time
    (b) The sum of all time spent in the ready queue
    (c) First CPU start time minus arrival time
    (d) Time spent performing disk I/O

32. What is turnaround time?

    (a) Completion time minus arrival time
    (b) First CPU start time minus arrival time
    (c) The length of one RR time slice
    (d) CPU burst time minus total waiting time

33. Which problem can arise with variable-size contiguous partitions?

    (a) All requests must waste exactly one full frame
    (b) No two adjacent holes can ever be merged
    (c) Every process must use the same-sized partition
    (d) External fragmentation can require compaction

34. With a fixed page size and PTE size, what happens to a full single-level page table when the virtual address space becomes larger?

    (a) It needs fewer entries
    (b) It needs more memory for page-table entries
    (c) The page offset becomes zero bits
    (d) It no longer requires address translation

35. Why can a multilevel page table save memory?

    (a) Unused lower-level page tables need not be allocated
    (b) All virtual pages become physically adjacent
    (c) Every PTE is kept inside a CPU register
    (d) Every second-level page table is always empty

36. What is a possible disadvantage of SJF when short jobs keep arriving?

    (a) It always has the largest average turnaround time
    (b) It must execute jobs in arrival order
    (c) A long waiting job can starve
    (d) It cannot run without equal-length bursts

37. Which statement about basic segmentation is NOT true?

    (a) A program can be divided into logical segments
    (b) Different segments can have different sizes
    (c) Each segment can use its own base and bound
    (d) All segments must be equal-sized units like pages

38. Which statement about paging is true?

    (a) A virtual page and a physical frame must differ in size
    (b) A page is a virtual-memory unit; a frame is a physical-memory unit
    (c) Smaller pages always produce a smaller full page table
    (d) Demand paging loads every page before execution

39. Which replacement policy requires future page references to make its ideal decisions?

    (a) FIFO
    (b) LRU
    (c) OPT
    (d) LFU

40. When does a demand-paging page fault occur?

    (a) When a referenced legal virtual page is not present in RAM
    (b) Whenever a resident page is read twice
    (c) Whenever an updated page is found in RAM
    (d) Whenever a translation is found in the TLB

41. Which policy rotates equal-priority ready jobs using a fixed time quantum?

    (a) Non-preemptive FIFO
    (b) Non-preemptive SJF
    (c) Round robin (RR)
    (d) OPT page replacement

Name _____ Student ID _____

Part II: Short Answer (10 points)

Answer all questions briefly in the spaces provided. (1 point each.)

1. Write the full names of MLFQ and STCF.

2. What is the difference between a process and a program?

3. Expand MMU and state its address-translation function.

4. Give two examples of CPU state saved during a context switch.

5. Give two types of information stored in a PCB, other than CPU register values.

6. What is an exited child process called before its living parent reaps its exit status?

7. Give one example each of an I/O-bound task and a CPU-bound task.

8. Why is longest-job-first usually a poor policy for reducing average turnaround time?

9. With base = 16 KiB and bound = 4 KiB, is virtual byte address 4200 valid? Why?

10. Expand TLB and explain why a TLB miss need not cause a page fault.

Name _____ Student ID _____

Part III: Written Questions (20 points)

Answer all three questions in the answer booklet. Show your work.

1. **(8 points)** Consider the following set of processes (jobs), where the run time of each process is given in milliseconds (ms):

Process	Run time (ms)	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

   The processes arrive in the order P_1, P_2, P_3, P_4, P_5, all at time 0. Assume zero context-switch overhead; break SJF ties by arrival order.

   (a) **(3 points)** Draw execution diagrams for FIFO, SJF, and RR scheduling with time slice = 1 ms.

   (b) **(2 points)** Find the turnaround time of each process under each policy.

   (c) **(2 points)** Find the response time of each process under each policy.

   (d) **(1 point)** Which policy gives (i) the minimum average turnaround time and (ii) the minimum average response time? Calculate both averages for all three policies.

2. **(4 points)** Explain how a commercial bank handles customers during rush hours with the concept of the Multi-level Feedback Queue (MLFQ), in your opinion. Describe the algorithms and policies used and provide suggestions on how to improve the process with the goal of optimizing the average turnaround time of customers.

3. **(8 points)** Use three initially empty frames and this page-reference string (20 references):

   7, 0, 1, 3, 0, 3, 2, 4, 2, 7, 0, 3, 2, 1, 2, 0, 1, 4, 0, 1

   Assume memory-access time is 10 ns and disk-access time is 10 ms. For EAT, treat 10 ms as the complete fault-path cost (including restart) and ignore other costs.

   (a) **(4.5 points)** Draw FIFO, LRU, and OPT frame tables; mark each reference as a hit or fault.

   (b) **(1.5 points)** Find each policy's total faults, total hits, and fault probability.

   (c) **(2 points)** Calculate the effective access time (EAT) of each policy in milliseconds.

Name _____ Student ID _____

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